

Paarth Saxena

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PROFILE

- **Innovation-Oriented** Professional with **5+ Years** of Experience in Programming
- Designed Efficient Processes for High-Speed, **Semi-Automated Backend Development** Leveraging **LLMs**
- Ventured into my **own Startup, Neuralperk**, to create a digital platform through which Individuals can Lend and Rent Computing Resources
- Transformed **Neuralperk** from a Concept to a **Fully Productionized Application**
- Created a **Standalone Python Library** for **Neuralperk**
- Partnered with DTU's Tech Team, Team Defiance Racing (TDR), to Develop a Self-Driving Racing Car for **National** and **International Competitions**, including **Formula Bharat**

EDUCATION

Delhi Technological University

Bachelor of Technology in Mathematics and Computing

Delhi, India

Aug. 2023 – May 2027

PROJECTS

Neuralperk | *Docker, Python, Redis, React.js, Electron Js* | [Link](#)

A **P2P platform** through which individuals can **Lend and Rent Computing Resources for GPU Workloads**.

- **Developed and Deployed Scalable Server**: Built a fully scalable server using a microservices architecture, ensuring flexibility and high performance.
- Created a **User-Friendly Desktop Application**: Developed a desktop application with Electron.js and React, featuring **auto-update functionality** for seamless interaction. The application installs and configures **WSL** during installation to enable **Docker container execution**.
- **Affordable and Efficient GPU Resource Platform**: Provided an **affordable, high-efficiency platform** for **PhD researchers and 3D artists** to complete **GPU-related workloads** at a lower cost than cloud-based solutions.
- Developed and Deployed a python Library named **Neuralperk** enabling customers to pass their models | [Link](#)

NN From Scratch | *C++, Python* |

Benchmarking FCN Implementation: TensorFlow vs Custom Solution

- **Implemented FCN in Python and C++**: Developed a Fully Convolutional Network (FCN) using Python with NumPy and custom neural network algorithms, and simultaneously implemented the same FCN in C++ with custom algorithms.
- **Conducted Speed Benchmarks**: Compared training times between TensorFlow-based FCN and custom implementations, demonstrating that the custom implementation outperformed TensorFlow by up to **20x in specific scenarios**.

Avatar | *Javascript, Python, MongoDB, LangChain* |

Used to Improve the Efficiency and Communication between Team Members

- Developed a **RAG (Retrieval Augmented Generation) System**: This system serves as a **shared knowledge base**, allowing team members to ask questions and gain better clarity on project-related information
- **Facilitated Knowledge Creation and Storage**: Empowered teams to effectively create and store knowledge, **streamlining communication** and **improving efficiency** within large teams.
- **Enabled Memory and Template Creation**: The ability to create custom memories and memory templates allows teams to develop specific chats or chatbots. This feature enables teams to interact and gain insights into both simple and complex project components they are working on.

EXPERIENCE

Founder

Jan 2024 – Feb 2025

Neuralperk

- **Conducted Market Research and Presentations**: Validated ideas through market research and discussions with PhD scholars, uncovering the challenges they face due to insufficient computing resources and identifying missed opportunities.
- **Closed International Clients**: Identified and addressed technical bottlenecks and opportunities in the 3D rendering space, securing international clients by offering tailored solutions.

Trainee

Aug 2023 – Jan 2024

Team Defiance Racing (TDR)

- Leveraging ROS and Linux for Racing Robot Simulation
- Learning About the Core concepts and Technology behind Implement a Self – Driving Race Car.

SKILLS

Languages: Python, C/C++, JavaScript, SQL

Technical Skills : Fullstack Development, Version Control (Git, Github), System Design, Problem Solving, Quick Prototyping and Iteration, Linux, Cloud Deployments, Docker, Testing and Debugging

Soft Skills: Analytical & Strategic Thinking, Innovation & Creativity, Collaboration & Communication

ACHIEVEMENTS

3rd Rank International in International Olympiad of Mathematics and was Awarded a Mobile Handset

₹50K Grant from **DTU Incubation Centre** for My Startup, **Neuralperk**

Finalist (Top 10) for Ideathon in IIITM out of **320** Registrations